Tutorial - 4

(1) $T = ml^2\dot{\theta}_1^2 + \frac{1}{2}ml^2\dot{\theta}_2^2 + ml^2\cos(\theta_1-\theta_2)\dot{\theta}_1\dot{\theta}_2$

$V = -2mgl\cos\theta_1 - mgl\cos\theta_2$

for eq^m pts:- $\frac{\partial V}{\partial \theta_1} = 0 \Rightarrow 2mgl\sin\theta_1 = 0 \Rightarrow \theta_1 = 0.$

$\frac{\partial V}{\partial \theta_2} = 0 \Rightarrow mgl\sin\theta_2 = 0 \Rightarrow \theta_2 = 0.$

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we analyse the pts in the region:

$\theta_1 = 0 + \eta_1$; $\theta_2 = 0 + \eta_2$; $\eta_1, \eta_2 << 1$.

$\Rightarrow \dot{\theta}_1 = \dot{\eta}_1$; $\Rightarrow \dot{\theta}_2 = \dot{\eta}_2$

$$\therefore T = \frac{1}{2} ml^2 \dot{\eta}_1^2 + \frac{1}{2} ml^2 \dot{\eta}_2^2 + ml^2 \underbrace{\cos(\eta_1 - \eta_2)}_{1} \dot{\eta}_1 \dot{\eta}_2$$

$[\eta_1 - \eta_2 \approx 0 \Rightarrow \cos(\eta_1 - \eta_2) = 1]$

$$V = -2mgl\cos\eta_1 - mgl\cos\eta_2$$
$$\approx -2mgl\left(1 - \frac{\eta_1^2}{2}\right) - mgl\left(1 - \frac{\eta_2^2}{2}\right)$$
$$= -3mgl + mgl\eta_1^2 + \frac{mgl}{2}\eta_2^2$$

$$\therefore M = \begin{pmatrix} 2ml^2 & ml^2 \\ ml^2 & ml^2 \end{pmatrix} ; V = \begin{pmatrix} 2mgl & 0 \\ 0 & mgl \end{pmatrix}$$

$\therefore |V - \omega^2 M| = 0$

$$\Rightarrow \begin{vmatrix} 2ml(g - \omega^2 l) & -\omega^2 ml^2 \\ -\omega^2 ml^2 & ml(g - \omega^2 l) \end{vmatrix} = 0$$

$$\Rightarrow 2m^2l^2(g - \omega^2 l)^2 - (\omega^2 ml^2)^2 = 0$$
$$\Rightarrow 2\cancel{m^2l^2}(g - \omega^2 l)^2 = \omega^4 \cancel{m^2 l^4} l^2$$

$$\Rightarrow 2g^2 + 2\omega^4 l^2 - 4\omega^2 gl = \omega^4 l^2$$
$$\Rightarrow \omega^4 l^2 - 4\omega^2 gl + 2g^2 = 0.$$

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$$\therefore \omega^2 = \frac{4gl \pm \sqrt{16g^2l^2 - 8g^2l^2}}{2l^2}$$

$$= \frac{2g}{l} \pm \sqrt{\frac{2g^2}{l^2}} = 2\frac{g}{l} \pm \sqrt{2}\frac{g}{l} = \frac{g}{l}(2 \pm \sqrt{2}).$$

$$\therefore \omega_1^2 = \frac{g}{l}(2+\sqrt{2}) \quad ; \quad \omega_2^2 = \frac{g}{l}(2-\sqrt{2}).$$

$$\therefore \underline{\omega_1^2 = \frac{g}{l}(2+\sqrt{2})} :$$

$$\begin{pmatrix} 2ml(g - 2g - \sqrt{2}g) & -mgl(2+\sqrt{2}) \\ -mgl(2+\sqrt{2}) & ml(g - 2g - \sqrt{2}g) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} +2(1+\sqrt{2}) & 2+\sqrt{2} \\ 2+\sqrt{2} & +(1+\sqrt{2}) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\Rightarrow (2+\sqrt{2})a_1 + (1+\sqrt{2})a_2 = 0.$$

$$\Rightarrow a_2 = -\left(\frac{2+\sqrt{2}}{1+\sqrt{2}}\right)a_1 = -\sqrt{2}a_1$$

$$\therefore \boxed{A_1 = \begin{pmatrix} 1 \\ -\sqrt{2} \end{pmatrix}} \text{ (Ans)}$$

$$\therefore \underline{\omega_2^2 = \frac{g}{l}(2-\sqrt{2})} :$$

$$\begin{pmatrix} \cancel{2ml}g(-1+\sqrt{2}) & -\cancel{mgl}(2-\sqrt{2}) \\ -\cancel{mgl}(2-\sqrt{2}) & \cancel{ml}g(-1+\sqrt{2}) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\Rightarrow \frac{(2-\sqrt{2})}{\sqrt{2}-1}a_1 = a_2 = \sqrt{2}a_1 \quad ; \quad \therefore \boxed{A_2 = \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}} \text{ (Ans)}$$

(2) $T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2$

$= m\dot{x}_1^2 + \frac{1}{2} m\dot{x}_2^2$

$V = 2kx_1^2 + \frac{1}{2}k(x_2 - x_1)^2 + kx_2^2$

$= 2kx_1^2 + \frac{k}{2}(x_1^2 + x_2^2 - 2x_1x_2) + kx_2^2$

$= \frac{1}{2}\left[5kx_1^2 + 3kx_2^2 - 2kx_1x_2\right]$

$\therefore M = \begin{pmatrix} 2m & 0 \\ 0 & m \end{pmatrix}$; $V = \begin{pmatrix} 5k & -k \\ -k & 3k \end{pmatrix}$

$\therefore |V - \omega^2 M| = 0.$

$\Rightarrow \begin{vmatrix} 5k - 2m\omega^2 & -k \\ -k & 3k - \omega^2 m \end{vmatrix} = 0$

$\Rightarrow (5k - 2\omega^2 m)(3k - \omega^2 m) - k^2 = 0$

$\cancel{\Rightarrow (4k - 2\omega^2 m)(}$

$\Rightarrow 15k^2 - 5\omega^2 mk - 6\omega^2 mk + 2m^2\omega^4 - k^2$

$\Rightarrow 2m^2\omega^4 - 11\omega^2 mk + 14k^2 = 0$

$\therefore \omega^2 = \frac{11mk \pm \sqrt{121m^2k^2 - 112m^2k^2}}{4m^2}$

$= \cancel{\frac{11k}{4m} \pm} \frac{(11 \pm 3)}{4}\frac{mk}{m^2} =$

$\therefore \omega_1^2 = \frac{7}{2}\frac{k}{m}$; $\omega_2^2 = \frac{2k}{m}$. (Ans)

$\omega_1^2 = \frac{7k}{2m}$:

$5k - 2m\omega_1^2 = 5k - 7k = -2k$

$3k - m\omega_1^2 = 3k - \frac{7k}{2} = -\frac{k}{2}$

$$\begin{pmatrix} -2k & -k \\ -k & -\frac{k}{2} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0.$$

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 1 & 1/2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0 \quad \Rightarrow \quad a_2 = -2a_1$$

$$\therefore \boxed{A_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}} \quad \text{(Ans)}$$

$\omega_2^2 = \frac{2k}{m}$:

$5k - 2m\omega_2^2 = k$; $3k - m\omega_2^2 = k$.

$$\begin{pmatrix} k & -k \\ -k & k \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0 \quad \Rightarrow \quad a_1 = a_2$$

$$\boxed{A_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}} \quad \text{(Ans)}$$